# Accessible Slides in LaTeX

https://github.com/rhstanton/accessible_LaTeX

Version 1.4

Richard Stanton

richard.stanton@berkeley.edu

UC Berkeley

March 17, 2026

# Introduction

- From April 2026, course materials must meet WCAG 2.1 Level AA accessibility standards
- Applies even to password-protected course sites (e.g., bCourses).
  - (Official campus guidance issued Jan 8, 2026)
- Many of us use LaTeX to create teaching materials—both slides and documents.
- Standard LaTeX (including Beamer) does not automatically generate accessible PDFs.
- The good news: The LaTeX Project team has developed solutions!

# What is this project?

- An experiment exploring accessible LaTeX documents using the LaTeX Tagging Project
  - Two templates: Articles (`article` class) and slides (`ltx-talk`)
  - Shows: What is shared and what differs
  - Contains: Examples with math, text, graphics, and tables
  - Scores: Slides score 100% with Ally and pass common accessibility validators (veraPDF, PAC, Acrobat).

# How to use it?

1. **Learn the common requirements:** Apply to any LaTeX document
2. **See class-specific needs:** Understand slides vs. articles
3. **Copy and adapt:** Use as templates for your documents
4. **Study the code:** Heavily commented for learning

- Available at: https://github.com/rhstanton/accessible_LaTeX

# Converting to accessible LaTeX: Overview

- **Your existing LaTeX skills transfer**: you're just adding accessibility features
- **The changes are minimal** and follow consistent patterns

- **Common requirements** (both slides & articles):
  - Add document metadata and enable PDF tagging
  - Tag images with alt text; tag table headers
  - Use accessible colors; compile with LuaLaTeX

- **Key difference:**
  - Articles: Keep your existing class! Just add accessibility features
  - Slides: Switch to `ltx-talk` (similar syntax, but recreate styling)

- Both require LaTeX kernel 2025-11-01 or later (see next slides for details)
- See "Getting started" slides at end for detailed steps
- LaTeX Tagging Project documentation

# TeX version requirements (both slides and articles)

- Minimum: TeX Live 2023 or later with all packages updated
- Critical: Must have LaTeX kernel 2025-11-01 or later
  - Update via TeX Live Manager (Windows) or TeX Live Utility (Mac)
  - Or use Overleaf Labs (see next slide)
- Will NOT work: TeX Live 2022 or earlier

# You **CAN** use Overleaf (both slides and articles)

- `ltx-talk` requires a very recent **LaTeX kernel** (2025-11-01 or later)
- This is available through Overleaf's Labs program (not standard Overleaf)

- **Using Overleaf (2 steps)**
  1. **Join Overleaf Labs**:
     - Visit https://www.overleaf.com/labs/participate
     - Opt in and enable "Rolling TeX Live releases"
  2. **Configure project**:
     - Set TeX Live version to "Rolling TeXLive (labs)" (bottom of list)
     - Set Compiler to LuaLaTeX

- **Resources**: https://docs.overleaf.com/writing-and-editing/creating-accessible-pdfs

# Common requirements (1/3): Document Metadata

- **REQUIRED for both slides AND articles**

- Every accessible document must begin with `\DocumentMetadata`
- Goes *before* `\documentclass`

```
\DocumentMetadata{
    pdfstandard=a-2u,        % PDF/A-2u format
    pdfstandard=ua-1,        % Add PDF/UA-1 conformance target
    pdfversion=1.7,          % Explicit PDF version
    lang=en-US,              % Language for screen readers
    tagging=on,              % Enable PDF tagging

    tagging-setup={
        math/alt/use,          % Keep math accessibility enabled
        math/mathml/AF=false,  % Do not embed MathML as PDF Associated Files
        math/tex/AF=false,     % Do not embed TeX source as Associated Files
        math/mathml/sources=   % Clear MathML source attachment list
    }
}
```

- Configures accessibility, math tagging, and validator-compatibility behavior in one place.

# Common requirements (2/3): Images, Tables and Colors

- **REQUIRED for both slides AND articles**

- **Images:** All images must include alt text descriptions
  - Allows screen readers to describe visual content
  - Even decorative images need marking
  - See dedicated “Figures” slide for examples

- **Tables:** Must specify which rows are headers
  - Helps screen readers navigate table structure
  - Essential for data accessibility
  - See dedicated “Tables” slide for examples

- **Accessible colors:** WCAG 2.1 requires 4.5:1 contrast ratio

```
\colorlet{AccessibleRed}{red!80!black}
\colorlet{AccessibleGreen}{green!40!black}
% Standard blue is fine as-is
```

# Common requirements (3/3): LuaLaTeX

- **REQUIRED for both slides AND articles**
- You must compile with LuaLaTeX (not pdfLaTeX or XeLaTeX)
- **Why LuaLaTeX?**
  1. **Automatic MathML:** Makes math accessible without manual work
  2. **Full tagging support:** Complete PDF accessibility features
  3. **Modern fonts:** Handles Unicode and OpenType fonts
- **How to switch:**
  - Command line: `lualatex filename.tex`
  - Most editors: Select “LuaLaTeX” from compiler menu

# The basics

- Slides are put inside a `frame` environment, just like in Beamer.
- So **existing source files don't need a lot of editing.**
- Here's some gratuitous $\mathit{math}$ for the accessibility checker.

```
\begin{frame}{The basics}
\begin{itemize}
\item Slides are put inside a \texttt{frame} environment, just like in Beamer.
\item So \textbf{\color{blue}existing source files don't need a lot of editing.}
\item Here's some gratuitous $\mathit{math}$ for the accessibility checker.
\end{itemize}
\end{frame}
```

- Note: I set the fonts in this file so that numbers, percent signs, and dollar signs look the same in both math and text mode (my pet Beamer peeve...)
  - Text: $1234567890%.
  - Math: $\$1234567890\%$.

# Figures

- Including figures is the same as in Beamer (e.g., using `\includegraphics`), but you need to provide a text description.

```
\includegraphics[height=.4\textheight,alt={A capybara}]{capybara.jpg}
```

# Tables

- Including tables is the same as in Beamer (e.g., using the `tabular` environment).
- Also need to specify header rows. Use `{1}` for 1 header row, `{1,2}` for 2 rows, etc.
- **Note:** This example may still trigger PAC's `Table header cell has no associated subcells` warning even when the source tagging is correct.

```
\tagpdfsetup{table/header-rows={1,2,3}}
\begin{tabular}{ccccrcccr}
\toprule
← 3 header rows
\midrule
← data rows
\bottomrule
\end{tabular}
```

| Payment date | Caplet expiry date | $DF_{pay}$ | Forward rate | Days to expiry | Days in accrual period | $T_{expiry}$ | $\Delta$ | Caplet |
|---|---|---|---|---|---|---|---|---|
| 2004/12/01 | – | 0.99550 | 0.01790 | 0 | 91 | 0.00000 | 0.25278 | – |
| 2005/03/01 | 2004/11/29 | 0.99008 | 0.02188 | 89 | 90 | 0.24384 | 0.25000 | 1,178.77 |
| 2005/06/01 | 2005/02/25 | 0.98401 | 0.02413 | 177 | 92 | 0.48493 | 0.25556 | 4,844.73 |
| 2005/09/01 | 2005/05/27 | 0.97733 | 0.02675 | 268 | 92 | 0.73425 | 0.25556 | 10,016.71 |

# Common pitfalls

- Confusing PDF tagging with PDF bookmarks
  - `tagging=on` creates structure tree for screen readers (required!)
  - PDF bookmarks (navigation pane) are separate and optional
  - Bookmarks do NOT generate automatically—need custom code
  - This template includes bookmark code; see preamble Part 2
- Forgetting alt text for images
  - Every `\includegraphics` needs an `alt={...}` parameter
  - For purely decorative images, use empty alt text: `alt={}`
- Not specifying table header rows
  - Add `\tagpdfsetup{table/header-rows={...}}` before each table
  - Use `{1}` for 1 header row, `{1,2}` for 2 header rows, etc.
- Insufficient color contrast
  - WCAG 2.1 requires 4.5:1 contrast ratio for normal text
  - Avoid light colors: `yellow`, `cyan` fail contrast requirements
  - Darken `red` and `green`: use `red!80!black`, `green!40!black`
  - Standard `blue` is fine and meets WCAG requirements
  - Test with a contrast checker: https://webaim.org/resources/contrastchecker/
- Using the wrong compiler
  - Make sure your editor is set to use LuaLaTeX, not pdfLaTeX
- Old TeX distribution
  - TeX Live 2022 or earlier won't work
  - Update packages using TeX Live Manager (Windows) or TeX Live Utility (Mac)

# What to ADD to existing documents

- **Converting existing LaTeX/Beamer files? Here's what to add:**

- **1. At the very top** (before `\documentclass`):

```
\DocumentMetadata{...tagging=on...}
```

- **2. In every** `\includegraphics`:

```
\includegraphics[alt={Description}]{file}
```

- **3. Before every table**:

```
\tagpdfsetup{table/header-rows={1}}
```

- **4. Change compiler to LuaLaTeX**

- **5. For slides only:** Change `\documentclass{beamer}` to `{ltx-talk}`
  - Keep your `\begin{frame}` environments as-is
  - Remove Beamer themes/templates; recreate styling with standard LaTeX

# Getting started: Common steps

1. **Setup environment:** Use Overleaf Labs OR install/update TeX Live locally
    - Overleaf: See earlier slides for Labs setup
    - Local: Update via TeX Live Manager/Utility
2. **Get templates:** Download from https://github.com/rhstanton/accessible_LaTeX
3. **Add accessibility features:**
    - Add `alt` text to images
    - Add `table/header-rows` to tables
4. **Set compiler to LuaLaTeX**
5. **Compile:**
    - Compile with LuaLaTeX
    - PDF file now scores 100% on Ally

# Getting started: What's different by document type

**Articles (using `article`, `report`, `book`, etc.):**

- Add `\DocumentMetadata` before `\documentclass`
- Include the `tagging-setup` block in `\DocumentMetadata` (as shown in this template)
- Switch to `fontspec` and `unicode-math` packages
- Keep your existing documentclass!

**Slides (migrating from Beamer):**

- Copy preamble from `accessible_slides.tex`
- Change `\documentclass{beamer}` to `\documentclass[frame-title-arg]{ltx-talk}`
- Remove Beamer themes/templates/colors
- Recreate styling using standard LaTeX/xcolor
- One-time preamble work—then reuse for all future talks!

- Questions? richard.stanton@berkeley.edu

# Validation workflow: use multiple checkers

- No single checker catches everything.
- Accessibility usability checks and strict PDF conformance checks are not identical.
- A practical workflow is to run more than one checker and compare findings.

- **Checkers used so far**
  - Ally (built into bCourses)
  - veraPDF (open-source PDF conformance validator, https://verapdf.org/)
  - PAC (PDF Accessibility Checker, https://pac.pdf-accessibility.org/)
  - Adobe Acrobat.

# Optional validator cleanup step: `strip_af.py`

- The metadata settings alone are enough to generate a 100% score on Ally.
- Stricter checkers (e.g., veraPDF) may still complain about /AF or embedded-file structures.
- To remove these and stop the complaints, run `strip_af.py`:

```
python strip_af.py build/accessible_slides.pdf
```

- Prerequisite: install the Python package pikepdf first: `python -m pip install pikepdf`.
- Then validate `build/accessible_slides.pdf` with `veraPDF`.

# Validator-specific notes

- **Acrobat: “Lbl and LBody – Failed” (validator bug)**
  - Appears around figure captions, section numbers, table captions
  - This is an Acrobat bug, not a source error – safe to ignore
  - The `<Lbl>` tag is correctly used for numbering
  - Acrobat’s checker incorrectly expects it only in lists
  - Reference: https://tex.stackexchange.com/a/709655/2388
- **PAC table warning (known behavior)**
  - PAC can report `Table header cell has no associated subcells` even when `table/header-rows` is correct
  - This behavior is documented by the LaTeX Tagging Project: https://github.com/latex3/tagging-project/issues/1056
- **When this warning appears**
  - First confirm source markup is correct (especially `table/header-rows`)
  - Keep the source tagging and document the validator warning
  - Report minimal reproductions to tool vendors when possible
- **Bottom line**
  - Use multiple validators (Ally, veraPDF, PAC, Acrobat) and compare findings
  - Treat one-tool warnings as signals to investigate, not proof of errors